Guided Step-by-Step Demo: Setting Up a Global External Application Load Balancer in Google Cloud

Overview & Objectives

In this hands-on lab demo, you will learn how to configure a Global External Application Load Balancer (Layer 7) in Google Cloud to front-end multiple regional Compute Engine VM backends serving HTTP web traffic.

By the end of this demo, you will be able to:

1. Create managed instance groups (MIGs) across different regions or zones.
2. Configure a Backend Service with a health check.
3. Set up a URL map and global forwarding rule with an anycast IP.
4. Test global traffic routing and failover.

Task 1: Create Instance Templates and Managed Instance Groups (MIGs)

1. Navigate to **Compute Engine > Instance templates** and click **Create instance template**.

    - **Name:** `web-server-template`
    - **Machine type:** `e2-micro`
    - **Boot disk:** Ubuntu 24.04 LTS
    - Expand **Management, security, disk, networking, sole tenancy** -> **Management** -> **Startup script**:

      ```
      #!/bin/bash

      apt-get update

      apt-get install -y apache2

      echo "Welcome from Regional Web Server - $(hostname)" >
      /var/www/html/index.html
      ```

systemctl restart apache2

- Click **Create**.

2. Navigate to **Compute Engine > Instance groups** and click **Create instance group**:
    - **Name:** `migs-us-east1`
    - **Instance group type:** Managed instance group (stateless)
    - **Instance template:** `web-server-template`
    - **Location:** Single zone (`us-east1-b`)
    - **Autoscaling:** Off (Fixed size = 2)
    - Click **Create**.

Task 2: Configure Firewall Rules for Health Checks

1. Navigate to **VPC network > Firewall** and click **Create firewall rule**.
    - **Name:** `allow-lb-health-check`
    - **Network:** `default`
    - **Targets:** All instances in the network
    - **Source IPv4 ranges:** `35.191.0.0/16` and `130.211.0.0/22` (Google's health check prober ranges)
    - **Protocols and ports:** Check *TCP* and specify port `80`.
    - Click **Create**.

Task 3: Create the External Application Load Balancer

1. In the Google Cloud Console search bar, type **Load balancing** and open the **Load balancing** page.
2. Click **Create load balancer**.
3. Under **Application Load Balancer (HTTP/HTTPS)**, select **Start configuration**.
4. Choose **From Internet to my VMs** (External) and **Global external Application Load Balancer**, then click **Continue**.
5. Name your load balancer: `global-web-alb`.

Configure Backend

1. Click **Backend configuration** > **Backend services & backend buckets** > **Create backend service**.
    - **Name:** `web-backend-service`

    - **Backend type:** Instance group
    - **Instance group:** `migs-us-east1`
    - **Port number:** `80`
    - **Balancing mode:** Utilization
2. **Health Check:** Create a new health check:
    - **Name:** `http-basic-check`
    - **Protocol:** HTTP | **Port:** 80 | **Path:** `/`
3. Click **Save** and **Create**.

Configure Routing & Frontend

1. Click **Routing rules**: Accept default path routing (all paths go to `web-backend-service`).
2. Click **Frontend configuration**:
    - **Protocol:** HTTP (or HTTPS with a managed certificate)
    - **IP version:** IPv4
    - **IP address:** Ephemeral (creates a global anycast IP)
3. Click **Create**.

Task 4: Test Your Load Balancer

1. Once the load balancer provisioning is complete (green checkmark), copy its assigned **IPv4 address**.
2. Open a new browser tab or run a curl command from your local terminal:

    curl http://<YOUR_LOAD_BALANCER_IP>

3. **Expected Result:** You will see the welcome message served by your backend instances.